quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

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Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain analytically siloed because each was processed with a different software configuration. We present `quantmsdiann`, an open-source Nextflow/nf-core workflow that parallelises `DIA-NN` across cloud and HPC infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported `DIA-NN` version (extensible to any release via a custom container) through Docker and Singularity, with no manual installation or dependency

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resolution, and natively handles all vendor formats, converting to mzML when needed (e.g. SCIEX). It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a `pmultiqc` quality-control report. The parallelisation design reanalyses a 2,300-run single-cell dataset in 2.4 hours on an HPC cluster of 300 nodes.

We benchmark `quantmsdiann` on the full ProteoBench DIA panel (five `DIA-NN` versions × four instrument modalities) and reanalyse public DIA deposits spanning bulk cell lines (NCI-60/PXD003539, ProCan/PXD030304) and a single-cell oocyte plexDIA dataset (MSV000093870). SDRF-driven reprocessing with current `DIA-NN` versions matches or exceeds the originally deposited results, and five cell-line cohorts are further combined into a pan-cohort under one invocation. Broader single-cell and spatial demonstrations are ongoing. `quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann`.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultiqc

1. Introduction

Data-independent acquisition (DIA) is now the dominant mass-spectrometry strategy for proteomics at scale, and `DIA-NN` [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm `DIA-NN`'s competitiveness for single-cell DIA workflows [4], and recent releases have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. Despite this analytical maturity, most `DIA-NN` users run the tool as a desktop application, which limits its scalability for large datasets. Deployments on HPC and cloud infrastructure remain artisanal, relying on shell scripts, ad-hoc directory conventions, and hand-curated sample sheets. As a result, the same raw data analysed by two laboratories, or with two `DIA-NN` releases, can yield non-overlapping quantification tables that are difficult to reconcile.

Beyond per-study analysis, the value of public proteomics archives such as PRIDE Archive [5] lies as much in systematic reanalysis as in original deposition. In 2024 we published bigbio/quantms [6], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data

Relationship Format (SDRF) [7] input on the nf-core framework [8]. quantms was, however, built around `DIA-NN` 1.8.1 with a fixed parameter set, and its DDA/DIA breadth made extending DIA-specific functionality difficult. New `DIA-NN` parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run `DIA-NN` invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of nf-core, the sensitivity of current `DIA-NN` releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present `quantmsdiann`, an independent SDRF-driven Nextflow pipeline dedicated to `DIA-NN` that extends DIA-specific features beyond what the original bigbio/quantms pipeline could accommodate. While `quantmsdiann` shares its SDRF-first design philosophy and parts of the configuration-generation tooling with quantms, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every `DIA-NN` version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard `DIA-NN` reports and MSstats input, bundling the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival. We demonstrate the pipeline along three axes: ProteoBench [9] benchmarks across four DIA modalities and five `DIA-NN` releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan) and a single-cell oocyte plexDIA dataset; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation.

2. Experimental Procedures

2.1. Pipeline implementation

`quantmsdiann` is implemented in Nextflow DSL2 ($\geq$25.10.4) and follows nf-core community standards [8] for pipeline structure, testing, documentation, and continuous integration. It supports both `DIA-NN` library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode with the UniProt human reference proteome [10] (UP000005640, *[release]*) and match-between-runs enabled at the cohort level. Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [11], pridepy [12], QPX, and the `wiffconverter` SCIEX reader) are pulled from public registries (BioContainers or `ghcr.io/bigbio`). `DIA-NN` is handled separately because its license restricts redistribution: only version 1.8.1 is publicly distributable and is pulled as the default from BioContainers (`docker.io/biocontainers/diann:v1.8.1_cv1`), whereas `DIA-NN` releases from 1.9 onward (1.9.2 through 2.5.0) cannot be redistributed and are not shipped as images. For those versions the pipeline provides build recipes in the companion `quantms-containers` repository (`https://github.com/bigbio/quantms-containers`), one per release; a user with a valid `DIA-NN` download builds the container locally, for example

```
cd diann-2.5.0 && docker build -t ghcr.io/bigbio/diann:2.5.0
```

. For the workflow to pick up a locally built image automatically, it must be tagged with the prefix `ghcr.io/bigbio/diann:` followed by the release tag (e.g. `ghcr.io/bigbio/diann:2.5.0`). The repository at `https://github.com/bigbio/quantmsdiann` is organised as a standard nf-core pipeline with main workflow (`main.nf`), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.1.0 (release codename “Sao Pablo”, 2026-05-08) was used for the analyses reported in this paper.

2.2. SDRF parsing and configuration generation

`quantmsdiann` accepts SDRF [7] as primary input. SDRF parsing is performed using the sdrf-pipelines Python package developed within the quantms project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. `DIA-NN` configuration generation extracts enzyme, fixed and variable modification, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings from the SDRF and writes a `DIA-NN` command-line configuration. Where SDRF columns are absent, `DIA-NN` defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and `DIA-NN` execution

Raw inputs are dispatched to format-specific handlers in the file-preparation subworkflow, each conversion or decompression running as an independent, parallelised Nextflow process. Thermo `.raw` files are read natively by `DIA-NN` $\geq$ 2.1.0 (which added Linux native-reader support) or converted to mzML with ThermoRawFileParser [11] otherwise (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (`https://github.com/bigbio/wiffconverter`). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX *export*

Per-run and per-cohort quality control reports are generated by `pmultiqc` [13], a proteomics extension of MultiQC [14] that summarises `DIA-NN`-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (`https://github.com/bigbio/qpx`), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [15] downstream analysis.

2.5. Datasets benchmarked

The benchmark and reanalysis datasets used in this study (ProteoBench DIA modules 5/7/9/10, the three per-cohort cell-line reanalyses PXD003539, PXD004701, PXD030304, the two additional pan-cohort datasets PXD041421 and PXD017199, and the PXD071075 scalability sweep) are summarised together with their instrument, acquisition mode, and deposition year in Additional file 1: Table S1. For each dataset, an SDRF was prepared from the original sample metadata and is provided as part of the Additional file 1 deposition. For each cell-line reanalysis cohort, the original published quantification was downloaded from the PRIDE deposition and used as the comparator, thresholded at 1% precursor-level FDR for both `quantmsdiann` and the original analyses. The conservative contaminant/decoy filter and the like-for-like-threshold rule applied to the original-paper headline counts are described in Additional file 1.

2.6. *Compute environment and benchmarking*

Runtime benchmarks were performed on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory allocations dispatched by Nextflow. Memory profiling was performed using the Nextflow trace report (`-with-trace`).

3. Results

3.1. *End-to-end throughput on archive-scale DIA cohorts*

A single SDRF-driven invocation processes archive-scale DIA cohorts within hours of wall-clock time on commodity HPC, with wall-clock time decreasing monotonically with cluster width over the tested range up to a practical knee near 200 nodes. On a large single-cell DIA deposit (PXD071075; ~2,300 raw files), increasing the Nextflow `executor.queueSize` from 10 to 300 cluster nodes reduced end-to-end wall-clock time from 37.7 h to 2.4 h (Fig. 1b). Scaling was monotonic across the sweep, with a practical knee near 200 nodes (2.6 h) beyond which the cohort-level consolidation step dominates and additional parallel workers yielded diminishing returns. Beyond the 200-node knee, gains in wall-clock time come at decreasing CPU-hour efficiency: scaling from 10 to 300 nodes reduces wall-clock by 15.5× for a 30× increase in cluster width, so users should choose cluster width by the desired turn-around time rather than the maximum-throughput operating point. Across ProteoBench benchmark datasets and our public cell-line cohorts, final-run wall-clock time tracked cohort size and instrument family without per-dataset workflow changes (Fig. 1c): the largest cohort in the panel, ProCan (PXD030304; 5,798 TripleTOF 6600 runs), completed in 4.8 h on the same configuration that finished the 6-run ProteoBench cohorts in under one hour. Peak memory during the cohort-level consolidation step stayed within the 64–128 GB available on commodity HPC nodes, so `quantmsdiann` does not require high-memory specialist hardware. Per-step wall-clock and resource envelopes are reported in Additional file 1: Figs. S1 and S2.

3.2. *`quantmsdiann` pipeline architecture*

`quantmsdiann` is implemented in Nextflow DSL2 [8] and follows nf-core community standards, distributing each computational step as a versioned Docker or Singularity/Apptainer container. The pipeline accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. From this input, `quantmsdiann` derives the analysis parameters required by `DIA-NN` automatically, removing a common source of analysis-script divergence between research groups. The workflow comprises four mandatory and three optional modules organised around the `DIA-NN` analysis core (Fig. 1a), alternating between per-file parallel stages (red; in-silico library generation when requested, preliminary calibration, individual search) and collective stages (green; empirical library assembly, final quantification, MSstats conversion, `pmultiqc` [13] quality-control reporting) operating on the full cohort.

3.2.1. Parallelisation strategy

Raw-file conversion and per-run quantification are dispatched as independent Nextflow tasks, enabling the throughput scaling reported in Section 3.1. Dataset-wide `DIA-NN` steps that require the full cohort (match-between-runs, normalisation, protein inference) run in a single consolidation task with tuned memory ceilings, avoiding out-of-memory failures common to single-machine `DIA-NN` runs on 1,000+ file inputs. Outputs include the standard `DIA-NN` report tables (`report.tsv`, `report.pr_matrix.tsv`, `report.pg_matrix.tsv`) alongside MSstats-formatted input and a `pmultiqc` [13] quality-control report. A harmonised QPX archive bundles the SDRF, identifications (PSM, peptide, protein-group, feature), MS metadata, ontology terms, and pipeline provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [15] downstream analysis.

3.2.2. Version flexibility and library handling

Two capabilities of this architecture are exercised directly by the experiments below (full mechanism in Experimental Procedures). First, the same SDRF can be processed against any of the five container-pinned `DIA-NN` releases (1.8.1–2.5.0) in a single invocation, selected at runtime and extensible to further releases via a custom container; this enables the cross-version reproducibility audit of Section 3.3 without re-engineering the workflow. Second, the library-free and user-supplied-spectral-library modes (Experimental Procedures) let the same workflow serve bulk cohorts and the single-cell and spatial experiments where matched empirical libraries are often unavailable.

3.3. ProteoBench benchmark across DIA-NN versions and instruments

To test whether quantmsdiann preserves DIA-NN's analytical performance under SDRF-driven orchestration, we processed every public ProteoBench DIA module through the workflow and compared the resulting identification counts against the ProteoBench public-submission snapshot (29 May 2026) [9]. Because quantmsdiann runs DIA-NN library-free (predicting the library in-silico from the FASTA, with no externally supplied experimental library), we restricted the community set for a like-for-like comparison to ProteoBench DIA-NN submissions that used the same predicted-from-FASTA library (predictors_library = DIANN); submissions that loaded a pre-existing experimental or user-curated spectral library search a different candidate space and are excluded from the headline comparison. The benchmark panel covers four instrument families and acquisition modes: Orbitrap Astral 2 Th DIA (*Module 7*), single-cell DIA on Orbitrap Astral (*Module 9*, PXD049412), timsTOF diaPASEF (*Module 5*, PXD062685), and SCIEX ZenoTOF 7600 (*Module 10*, PXD070049). For each module we ran the workflow against five DIA-NN releases (1.8.1, 2.1.0, 2.2.0, 2.3.2, and 2.5.0) in a single invocation, with all five versions parametrised from the same SDRF (Fig. 2).

Precursor counts (quantified in at least one replicate) grew monotonically with DIA-NN version on every module, rising by 14% on Module 7 (105,793 → 120,973 from 1.8.1 to 2.5.0), 14% on Module 9 (38,218 → 43,761), 27% on Module 5 (91,075 → 115,735), and 17% on Module 10 (85,847 → 100,524). Protein-group counts at 1% FDR increased over the same range (endpoint gains of 7% / 8% / 14% / 7% respectively), though not strictly monotonically: on each module the protein count dips by 1–2% at an intermediate version before the newest release recovers the headline gain. Reproducibility-filtered precursor counts (≥3 replicates) can likewise be non-monotonic — on the ZenoTOF module (PXD070049), for instance, DIA-NN 2.5.0 yields the most total precursors but slightly fewer reproducible ones than 2.3.2. This monotonic version-progress trend is consistent with what the broader ProteoBench community reports across DIA-NN releases [9], indicating that the SDRF wrapper introduces no version-coupling artefacts. Quantification accuracy, by contrast, is essentially invariant across DIA-NN versions. On every module the measured per-species $\log_2$ fold-changes recover the expected HYE ratios—human at unity, yeast and *E. coli* near their design ratios, with the mild attenuation at the extreme *E. coli* ratio that is characteristic of DIA quantification—and the five DIA-NN releases overlie one another at every species (Fig. 2b). The version-to-version shift in median quantification error $|\epsilon|$ ($\sim$0.01 $\log_2$, $\approx$0.5–1% in linear fold-change) is far smaller than the per-precursor spread, so the choice of `DIA-NN` release is best driven by the identification gains in panel (a) rather than by accuracy. On the two modules for which the snapshot contained predicted-library `DIA-NN` submissions—Orbitrap Astral (Module 7, $n = 15$) and timsTOF diaPASEF (Module 5, $n = 8$)—`quantmsdiann`'s five versions fall within the community median-$|\epsilon|$ distribution, at or below its median (Fig. 2c); the single-cell and ZenoTOF modules carried no predicted-library community submission. Parameter-matched per-submission comparisons are provided in Additional file 1: Fig. S3.

3.4. Independent reanalyses of public DIA deposits recover comparable or deeper coverage than the original quantifications

To evaluate `quantmsdiann` on production-scale public deposits, we reanalysed independent DIA cohorts—spanning bulk cell lines and single cells—that had been processed and published with different DIA engines (Fig. 3). For each cohort the same raw files were retrieved from PRIDE Archive [5] (or MassIVE, for the single-cell plexDIA cohort), parsed through an SDRF generated from the original sample annotations, and run through `quantmsdiann` with current `DIA-NN` releases.

On the **NCI-60** panel (PXD003539, Guo *et al.* 2019 [16]; originally analysed with OpenSWATH against the Guo assay library), `quantmsdiann` recovered 95,633 peptides at 1% FDR against 40,592 in the original submission (+135%), with comparable protein-group counts (6,747 versus 6,556; Fig. 3a). Because the original used OpenSWATH with a fixed assay library while `quantmsdiann` runs `DIA-NN` library-free, this gain reflects both the SDRF-driven orchestration and the change of analysis engine; a like-for-like `DIA-NN`-on-original-library control is part of ongoing work. On the **single-cell oocyte plexDIA** cohort (MSV000093870, Galatidou *et al.* 2024 [17]; mTRAQ 3-channel, Q Exactive)—the first plexDIA dataset processed through `quantmsdiann`—the workflow quantified 2,904 protein groups from the same raw data, versus 2,122 in the original analysis (+37%). At the UniProt-accession level it recovered 95% of the originally reported proteins (2,013 of 2,122) and detected 1,539 further accessions, at comparable per-cell median depth (1,736 versus 1,784; Fig. 3b). On the **ProCan**-DepMapSanger 949-cell-line pan-cancer panel (PXD030304, Gonçalves *et al.* 2022 [18]; originally analysed with `DIA-NN` 1.7.x), `quantmsdiann` recovered 8,921 protein groups at 1% FDR against 8,498 originally (+5%) and 7,990 protein groups requiring ≥ 2 unique peptides versus 6,692 (+19%; Fig. 3c). Both the original 8,498 protein-group headline (Global.Q.Value ≤ 0.01, no per-protein peptide threshold) and the stringent 6,692 count (same filter plus a ≥ 2 supporting-peptide requirement) are reported in the ProCan paper itself; the `quantmsdiann` numbers are computed under the matching filters and the same conservative contaminant/decoy filter, so each side of the comparison is like-for-like. A fourth, bulk cohort already analysed with a current DIA engine—the Sun *et al.* 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 lines [19])—yielded near-parity (6,146 versus 6,091 protein groups, +0.9%; Additional file 1), as expected when the original analysis already used a modern engine. Across these cohorts the gain was largest for deposits processed with older or non-`DIA-NN` engines (NCI-60 OpenSWATH, ProCan 1.7.x) and for the single-cell plexDIA cohort, and smallest for the cohort already analysed with a current DIA engine. Per-condition completeness, peptide-per-protein distributions, gene-versus-accession overlap and per-cell-line missingness for each cohort are provided as Additional file 1: Figs. S5–S13.

3.5. Pan-cohort of five public DIA cell-line reanalyses

To illustrate the value of running the same SDRF-driven workflow across many independent deposits, we combined the three reanalysed cohorts above with two additional cell-line DIA panels (PXD041421, the MultiPro batch-effect testbed, Wang *et al.* 2023 [20]; PXD017199, Tognetti *et al.* 2021 [21], 67 breast-cancer lines) into a single `quantmsdiann` invocation and assembled a pan-cohort view of per-cohort yield and per-tissue coverage (Fig. 4). Per-cohort headline counts (Fig. 4a) confirm that `quantmsdiann` matches or exceeds the original published count on each cohort. Across the five cohorts the reanalysis spans 12,229 unique UniProt accessions, of which 5,144 (42.1%) are detected in all five—a pan-cohort core proteome—and a further 1,334 (10.9%) in exactly four; these overlap counts are tabulated in the combined per-cohort counts provided with the analysis code.

Tissue-level coverage across the combined panel spans 27 Cellosaurus-derived tissue categories (plus the catch-all "Other tissue" and "Healthy (Non-cancer)" bins; Fig. 4b,c): lung, breast, haematopoietic-and-lymphoid, skin, and central nervous system together account for 59% of all cell-line entries in the pan-cohort. Because every count comes from running a single SDRF-parameterised `quantmsdiann` invocation rather than reconciling per-study quantifications, the resulting protein matrix is queryable from the published

QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation. The consolidated SDRF and per-cohort `quantmsdiann` configurations underpinning this pan-cohort run will be deposited on Zenodo before publication, so that the full input–configuration–output triple can be re-executed by third parties.

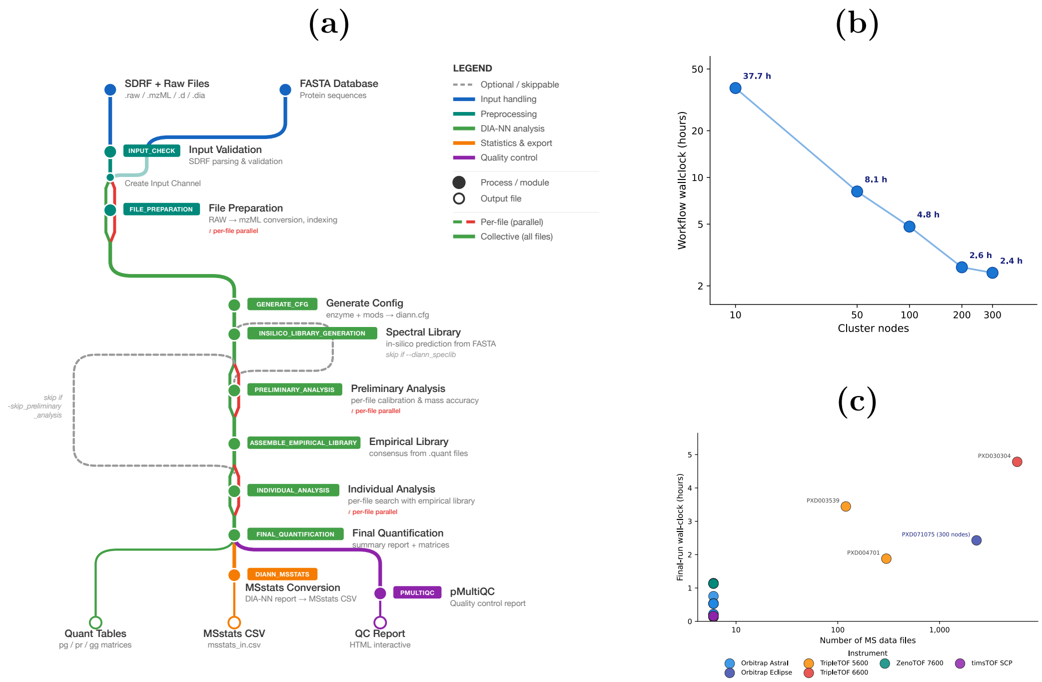

Figure 1: **`quantmsdiann` pipeline architecture and runtime scaling.** (**a**) Subway diagram of the `quantmsdiann` workflow. Mandatory modules are shown as filled circles, optional/skippable modules as open circles; per-file parallel stages are marked in red, collective cohort stages in green. The pipeline consumes an SDRF and a FASTA reference, optionally converts vendor raw files to mzML, generates an in-silico or empirical spectral library, runs per-file `DIA-NN` analyses in parallel, consolidates results in a single cohort-level final quantification, and emits standard `DIA-NN` tables, an MSstats input, a QPX Parquet archive, and a `pmultiqc` quality-control report. (**b**) Workflow wall-clock time versus Nextflow `executor.queueSize` (log-scaled cluster nodes; ticks at 10, 50, 100, 200, 300) on the PXD071075 single-cell cohort (~2,300 runs); wall-clock drops from 37.7 h at 10 nodes to 2.4 h at 300 nodes, with the practical knee near 200 nodes. (**c**) Final-run wall-clock versus number of MS data files across ProteoBench benchmark and cell-line cohorts, coloured by instrument family; the PXD071075 production run (300 nodes) is annotated.

(a)

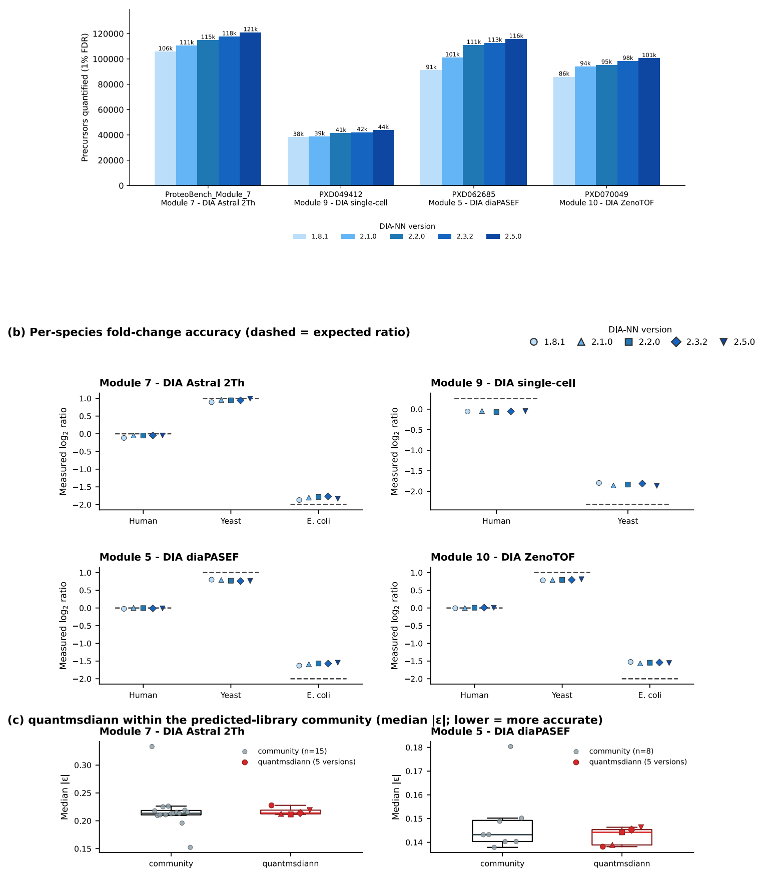

Figure 2: **quantmsdiann on ProteoBench across five DIA-NN releases and four DIA modalities** (Orbitrap Astral, Module 7; single-cell, Module 9/PXD049412; timsTOF diaPASEF, Module 5/PXD062685; SCIEX ZenoTOF, Module 10/PXD070049). (**a**) Precursors at 1% FDR by DIA-NN version (1.8.1–2.5.0; light to dark blue) from one quantmsdiann invocation; precursor counts grow monotonically with version (e.g. Module 5 91,075 → 115,735). (**b**) Per-species quantification accuracy: measured $\log_2$ fold-change for each HYE species against the ProteoBench-expected ratio (dashed), with all five DIA-NN versions overlaid (light to dark). The versions cluster tightly on every species and module — quantification accuracy is essentially invariant across DIA-NN releases, the version-to-version median $|\epsilon|$ shift ($\sim$0.01 $\log_2$) being far smaller than the per-precursor spread; the mild attenuation at the extreme *E. coli* ratio is the expected DIA ratio-compression. (**c**) For the two modules with predicted-from-FASTA (predictors_library = DIANN) DIA-NN community submissions (snapshot 29 May 2026), quantmsdiann's five versions (red) sit within the community median-$|\epsilon|$ distribution (grey; box = IQR). The single-cell and ZenoTOF modules had no predicted-library community submission. See Additional file 1: Fig. S3 for parameter-matched per-submission comparisons.

(a) PXD003539 — 2019, NCI-60 **(b) MSV000093870 — 2024, plexDIA** **(c) PXD030304 — 2022, ProCan**

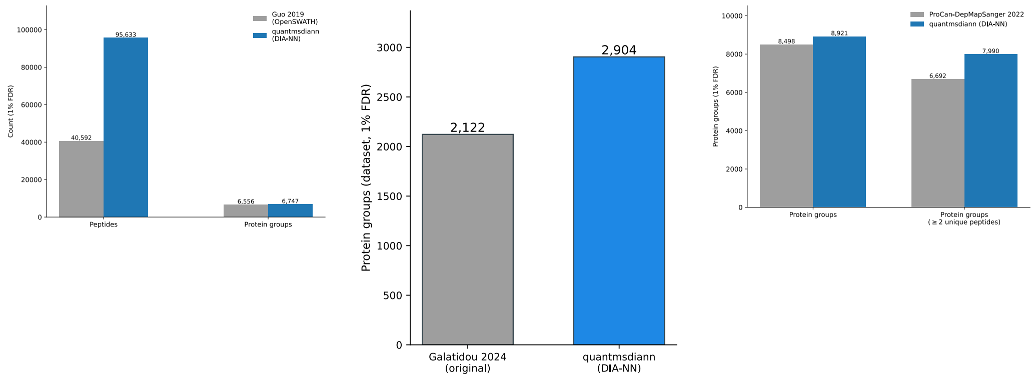

Figure 3: **Independent reanalysis of public DIA datasets, spanning bulk cell lines and single cells, recovers as many or more identifications than the originally deposited quantifications.** Each panel compares the originally published peptide and/or protein-group counts (grey) against the `quantmsdiann` reanalysis at 1% FDR (blue) using the same raw data. (**a**) Guo *et al.*, 2019 NCI-60 OpenSWATH analysis (PXD003539, original engine = OpenSWATH with the Guo 2019 assay library): `quantmsdiann` recovers 95,633 peptides versus 40,592 originally (+135%) and a comparable 6,747 protein groups versus 6,556. Because `quantmsdiann` runs `DIA-NN` in library-free mode while the original ran OpenSWATH against a fixed assay library, this gain reflects both SDRF-driven orchestration and the change of analysis engine; a like-for-like `DIA-NN`-on-original-library control is part of ongoing work. (**b**) Galatidou *et al.*, 2024 single-cell oocyte plexDIA (MSV000093870; mTRAQ 3-channel, Q Exactive): from the same raw data `quantmsdiann` quantifies 2,904 protein groups versus 2,122 in the original analysis (+37%); at the UniProt-accession level it recovers 95% of the originally reported proteins (2,013 of 2,122) and detects 1,539 further accessions, at comparable per-cell median depth (1,736 versus 1,784; full per-cell, total and accession-overlap comparison in Additional file 1). (**c**) ProCan-DepMapSanger 2022 pan-cancer cell-line panel (PXD030304, 949 lines): `quantmsdiann` recovers 8,921 protein groups at 1% FDR versus 8,498 originally (+5%), and 7,990 versus 6,692 after requiring ≥ 2 unique peptides per protein (+19%). Both the original 8,498 headline (Global.Q.Value ≤ 0.01) and the stringent 6,692 count (Global.Q.Value ≤ 0.01 plus ≥ 2 supporting peptides) are reported in the ProCan-DepMapSanger paper itself; the `quantmsdiann` counts are computed under the matching filters, so each side of the comparison is like-for-like. Per-condition completeness, peptide-per-protein distributions and gene- versus accession-level overlap are shown in Additional file 1: Figs. S5–S13.

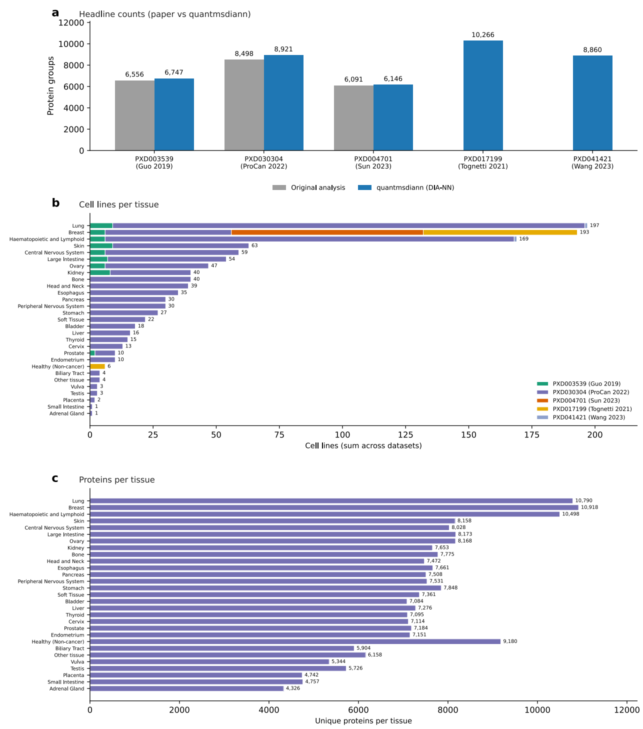

Figure 4: **Pan-cohort of five public DIA cell-line reanalyses by `quantmsdiann`.** Combines PXD003539 (NCI-60, Guo 2019), PXD030304 (ProCan-DepMapSanger, 949 lines), PXD004701 (Sun 2023, 76 breast cancer lines), PXD041421 (Wang 2023), and PXD017199 (Tognetti 2021, 67 breast cancer lines), reanalysed under a single `quantmsdiann` invocation. (**a**) Per-cohort headline protein-group counts: the originally published count (grey) and the `quantmsdiann` reanalysis (blue); PXD017199 and PXD041421 have no published DIA headline, so only the `quantmsdiann` bar is shown. (**b**) Cell lines per pan-cancer tissue, stacked by contributing cohort. (**c**) Unique UniProt proteins detected per tissue across the combined panel.

4. Discussion

`quantmsdiann` automates the path from a deposited DIA dataset to a reproducible quantification table, at a scale that supports systematic reanalysis of public DIA archives. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is interoperable with the wider proteomics ecosystem [7]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container so that reanalysed results can be archived (e.g. on Zenodo) and re-queried alongside the original raw data. By focusing on `DIA-NN` rather than the multi-engine scope of the related bigbio/quantms pipeline [6], `quantmsdiann` achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file `DIA-NN` invocation rather than across-engine comparison.

The demonstrations we present stress the workflow along three orthogonal axes. The ProteoBench benchmark (Fig. 2) tests analytical fidelity across four instrument modalities and five `DIA-NN` releases. The independent reanalyses (Fig. 3) test recovery against published per-study quantifications across bulk cell lines and single cells, with results ranging from near-parity on a cohort already analysed with a current DIA engine (Sun 2023; Additional file 1) to substantial recovery gains on the NCI-60 OpenSWATH panel, the ProCan `DIA-NN` 1.7.x panel, and the single-cell oocyte plexDIA cohort. The pan-cohort of five public DIA cell-line reanalyses (Fig. 4) tests archive-scale throughput on five concurrently reanalysed deposits. The oocyte plexDIA reanalysis (MSV000093870) is, to our knowledge, the first plexDIA dataset processed through an nf-core-style SDRF-driven workflow, and recovers 95% of the originally reported proteins while detecting 1,539 further UniProt accessions from the same raw data. This pattern suggests that the value of DIA reanalysis is real but time-limited—largest for deposits originally analysed with older or non-`DIA-NN` engines and shrinking toward parity for recently processed cohorts—a regime an SDRF-driven workflow is suited to address as new `DIA-NN` versions ship. Broader single-cell DIA and laser-microdissected spatial demonstrations are part of the development roadmap; the workflow already supports these acquisition modes (Fig. 1a, library-handling branch) and SDRFs for the Brunner [2], Derks [22], Leduc [23], Wang [4], Mund [24] and Makhmut [25] datasets are in preparation.

The cross-`DIA-NN`-version reproducibility view enabled by `quantmsdiann` is, to our knowledge, the first pipeline-level treatment of this question for

DIA-NN. We expect this to become increasingly important as DIA-NN evolves: a deposited single-cell DIA dataset analysed with DIA-NN 1.8 today and re-analysed with DIA-NN 2.x in two years should not produce different biological conclusions for the same data and parameters. In current practice such re-analyses are rarely performed, because the bespoke per-study scripts make them prohibitively expensive. quantmsdiann's container-pinned, version-parametrised execution reduces this kind of audit to a single workflow invocation.

quantmsdiann has several limitations. First, quantmsdiann does not currently include batch correction, cell-type assignment, or differential-abundance modules; these are best handled in downstream R or Python packages such as scp [26], which can read the QPX or MSstats outputs directly. Second, library generation for plexDIA channel deconvolution remains internal to DIA-NN, and quantmsdiann inherits any limitations of that step. Third, the QPX format will continue to evolve as community feedback accumulates; backward compatibility is maintained through schema versioning. Fourth, beyond the single-cell oocyte plexDIA reanalysis presented here, broader single-cell and laser-microdissected spatial DIA demonstrations are part of the development roadmap rather than evaluated in depth in this work.

Finally, we note that quantmsdiann is complementary to, rather than competing with, existing single-cell DIA workflows. The benchmarking work of Wang *et al.* [4] compared multiple search engines and rescoring strategies; quantmsdiann adopts the broadly competitive DIA-NN engine and focuses on the orthogonal problem of orchestrating that engine reproducibly at scale. Similarly, earlier nf-core DIA efforts such as DIAproteomics [27] and, more recently, the MHCquant2 pipeline [28] address analogous orchestration problems for DIA and immunopeptidomics. Together, these contributions extend the nf-core proteomics ecosystem to cover the major MS-based proteomics modalities behind a common SDRF-driven interface.

Declarations

Ethics approval and consent to participate

Not applicable. All datasets reanalysed in this study were obtained from public proteomics repositories with ethics approval and informed consent from the original depositing studies.

Consent for publication

Not applicable.

Data availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Scalability experiment: PRIDE PXD071075. ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Additional file 1: Table S1. Per-process runtime and resource envelopes (Figs. S1, S2) and detailed ProteoBench supplementary panels (Figs. S3, S4) are provided as Additional file 1.

Code availability

`quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann` under the MIT license; documentation is hosted at `https://quantmsdiann.quantms.org/`. The pipeline is implemented in Nextflow DSL2 ($\geq$25.10.4) and packages each step as a versioned Docker, Singularity, or Apptainer container image. `DIA-NN` 1.8.1 is distributed publicly via BioContainers; because the `DIA-NN` license restricts redistribution of later releases, containers for `DIA-NN` 1.9 and above are built locally from the recipes in `https://github.com/bigbio/quantms-containers` (see Experimental Procedures). Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at `https://github.com/bigbio/quantmsdiann-paper` (this repository).

Competing interests

[VD is the original author of DIA-NN. Other authors declare no competing interests.]

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Authors' contributions

[QXY, YS and CD developed the pipeline. AL and HW implemented the SDRF parsing and QPX export. JN and FC contributed the spatial-proteomics datasets and analysis. VD provided DIA-NN expertise and guidance on cross-version reproducibility. MB and YPR conceived and supervised the project. QXY and YPR wrote the manuscript with input from all authors. All authors read and approved the final manuscript.]

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